MPC-MAP Assignment 3

Week 4: Particle Filter

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Task 1 – Prediction

The particle filter represents the belief about the robot's position as a set of 1000 weighted samples (particles), each with a state $\mathbf{x} = [x, y, \theta]$. At the start, particles are initialized uniformly across the map.

The function `predict_pose.m` applies the differential-drive motion model with noise $\mathbf{Q}$ applied to the wheel velocities

Task 2 – Correction

For each particle, `compute_lidar_measurement.m` computes the expected lidar distances by casting 8 rays from the particle position and finding the nearest wall intersection. Beams that miss all walls return `inf`.

The weight of each particle is then set proportional to the likelihood of the measurement given the particle's state, using normal distribution.

Task 3 – Resampling

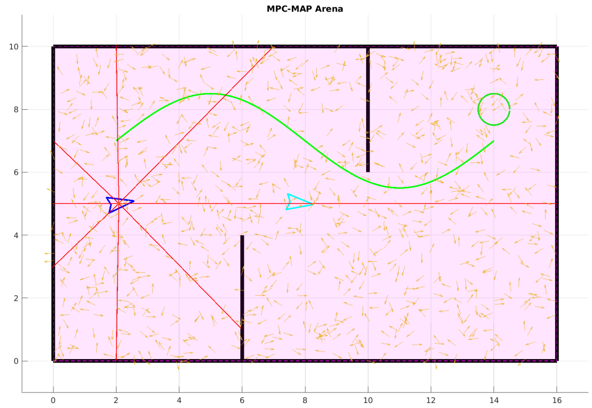
In the resampling step, a new set of particles is drawn from the current set with probability given by the weights. High-weight particles are duplicated, low-weight ones are discarded. This increases particle density in more probable regions of the state space.

Low variance systematic resampling is used: a single random offset $\tilde{u} \in [0, 1/N)$ is drawn and particles are selected at evenly spaced points $u_k = (k - 1 + \tilde{u})/N$ along the cumulative weight distribution.

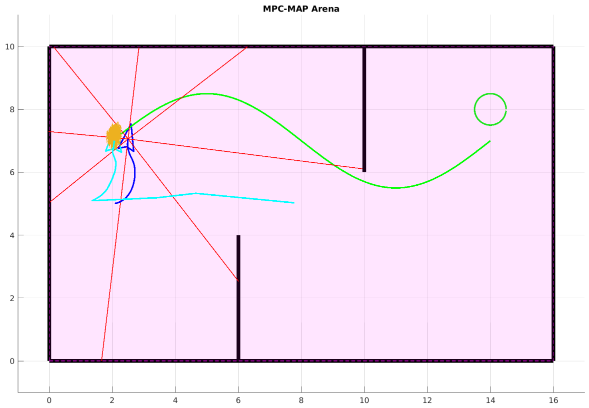
After resampling, small Gaussian noise is added to each particle ($\pm 0.02\,\mathrm{m}$ position, $\pm 0.01\,\mathrm{rad}$ heading) to reduce particle degeneracy. Without this, particles lose diversity after many resampling steps, which degrades tracking accuracy.

Task 4 – Localization

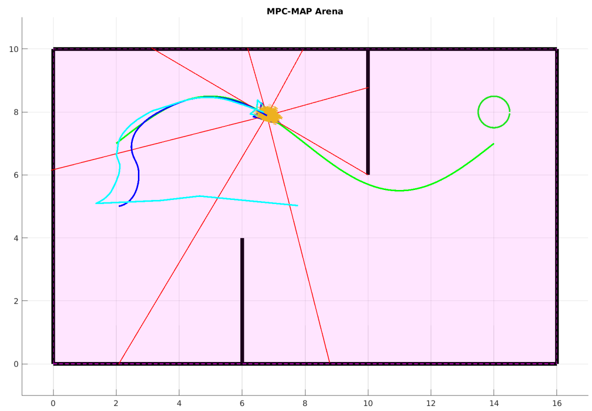
The full particle filter loop (prediction, correction, resampling) runs every time step. The implementation is in `update_particle_filter.m`. The estimated pose is the mean of all particles. The robot follows a sine-wave path on the `custom_1` map using the motion controller from Week 3.



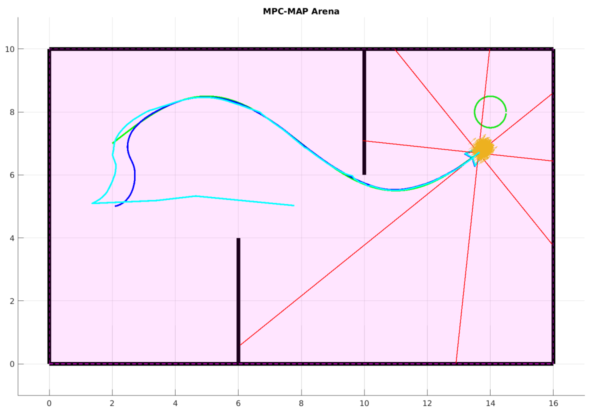
(a) Step 1



(b) Step 30



(c) Step 100



(d) Step 240

Figure 1: Particle filter convergence on `custom_1`.